Natural Language Understanding

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Problem 1: Word Embeddings

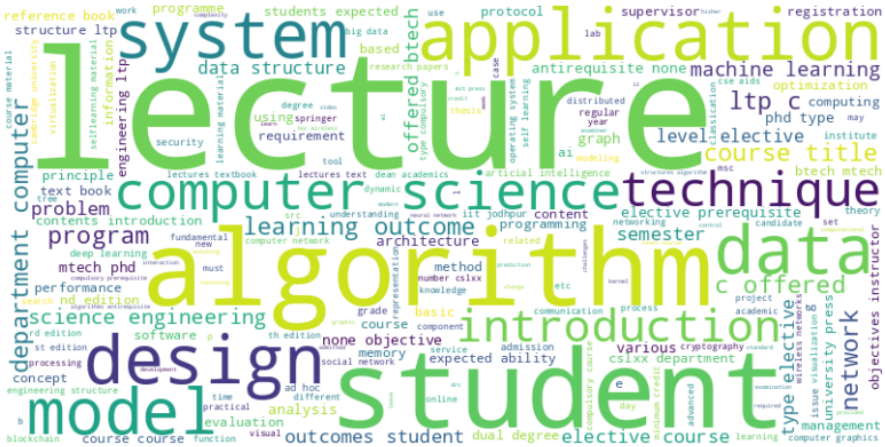
1 Task 1: Dataset Preparation

The dataset was constructed by scraping textual data from seven distinct sources related to IIT Jodhpur to ensure a diverse domain-specific corpus: the B.Tech programs page, the CSE department page, the Primary Health Center (PHC) page, the official Academic Regulations PDFs, the Courses offered by the CSE department, the Central Research Facility page, and the CSE Faculty Members page.
The raw text was preprocessed using NLTK and regular expressions. The pipeline included:

- Removal of HTML/PDF boilerplate and formatting artifacts.
- Conversion to lowercase.
- Removal of punctuation and non-alphabetic characters to prevent numerical noise from PDF page numbers.
- Tokenization and removal of standard English stop words.

Dataset Statistics

- **Total Documents:** 7
- **Total Tokens:** 33386
- **Vocabulary Size:** 5671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modeling
structure algorithm
msc
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theory
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networking
control
kernel
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Target Word	Top 5 Nearest Neighbors (with Cosine Similarity)
research	contribute (0.59), mobisys (0.54), papers (0.54), multidisciplinary (0.53)
student	students (0.65), heshe (0.65), indicated (0.62), assistantship (0.60)
phd	btech (0.66), mtech (0.64), msc (0.57), specialization (0.54)
examination	makeup (0.66), comprehensive (0.64), perform (0.64), attempt (0.62)

Table 2: Top Nearest Neighbors demonstrating domain-specific clustering.

3.2 Analogy Experiments & Qualitative Discussion

Analogy (A : B :: C : ?)	Predicted Word (Score)
BTech : Bachelors :: MTech : ?	**masters** (0.6337)
Machine : Learning :: Artificial : ?	**bioscience** (0.5093)
Data : Structures :: Operating : ?	**jpeg** (0.4617)

Table 3: Vector Arithmetic Analogy Results

Qualitative Discussion of Results:

1. **Domain-Specific Context over General English:** The model performed exceptionally well at mapping the bureaucratic reality of the institute. For instance, the neighbors for "student" heavily reflect the Academic Regulations corpus, associating it with formal handbook terms like "heshe" and "assistantship". Similarly, "examination" perfectly clusters with formal academic modifiers like "makeup" (makeup exams) and "comprehensive" (comprehensive exams).

2. **Degree Hierarchy:** The model successfully mapped the geometric relationships between academic degrees, flawlessly predicting `masters` when prompted with the difference between a B.Tech and a Bachelor's degree.

3. **Data Sparsity in Analogies:** Complex vector arithmetic requires massive, globally balanced datasets. Because our corpus is highly localized, the vector space is too sparse to resolve certain geometric equations reliably. For example, predicting `bioscience` instead of `intelligence`, or `jpeg` instead of `systems`, reflects how the model falls back on nearest-neighbor proximity when true vector subtraction isn't fully supported by the dataset's size.

4 Task 4: Visualization

Dimensionality reduction was applied using both Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbor Embedding (t-SNE) to project the 100-dimensional embeddings into 2D space. Figure 2 and Figure 3 compare the clustering behavior of CBOW and Skip-gram using the exact same hyperparameters ($d = 100, w = 5, n = 10$).

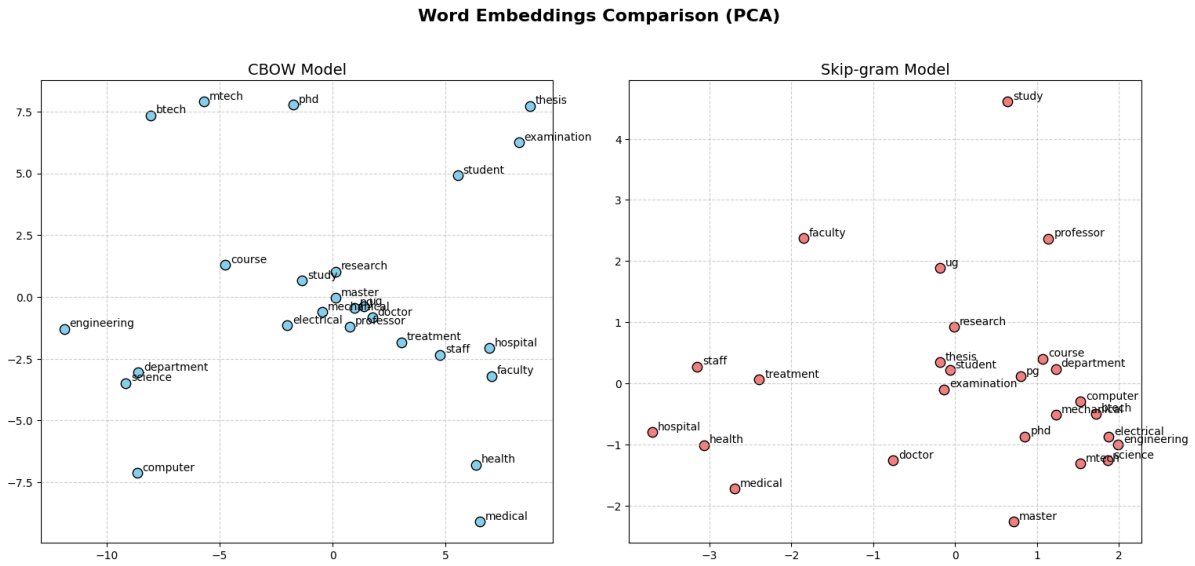

Figure 2: PCA Projection: Demonstrates the global geometric spread of CBOW (Left) versus Skip-gram (Right).

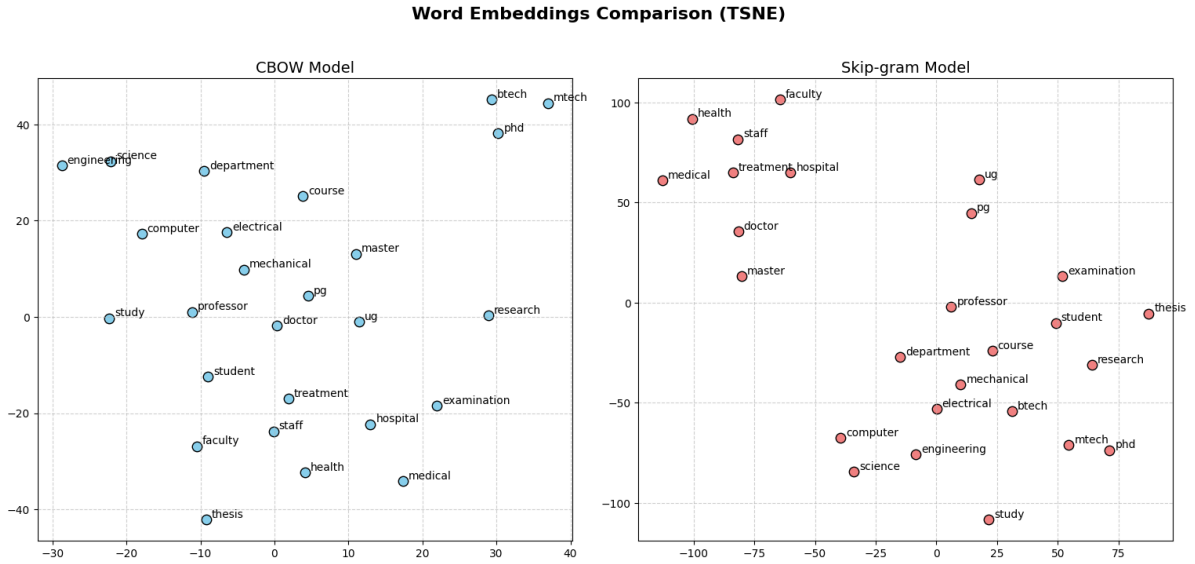

Figure 3: t-SNE Projection: Highlights the local, non-linear clustering behavior of CBOW (Left) versus Skip-gram (Right).

Interpretation of Clustering Behavior:

- **CBOW:** Because CBOW predicts a target word by averaging its context, it smooths over fine-grained details and aggressively groups words that share environments. In both the PCA and t-SNE plots, the distinct domains of the corpus (the Medical terms, the Engineering branches, and the Degree hierarchy) are packed into incredibly dense, isolated clusters.

- **Skip-gram:** Skip-gram uses a single target word to predict context, preserving much more individual semantic nuance. In the Skip-gram plots, the clusters still exist (e.g., the medical terms are grouped together), but they are significantly more dispersed. The model recognizes the relationship but refuses to crush the words into a single point, acknowledging that terms like "hospital" and "doctor" operate differently within a sentence structure.

Problem 2: Character Level Name Generation

1 Task 1: Model Implementation

Three sequence models were implemented from scratch using fundamental linear transformations to compare their ability to generate Indian names.

1.0.1 Architecture Descriptions

1. **Vanilla RNN:** A standard recurrent architecture where the hidden state h_t is updated at each step using the current input x_t and the previous state h_{t-1} via a $tanh$ activation. It is the most memory-efficient but prone to vanishing gradients.
2. **Bidirectional LSTM (BLSTM):** A more complex architecture utilizing four gated units (input, forget, output, and cell gates) to manage long-term dependencies. The bidirectional variant processes sequences in both forward and backward directions, though for step-by-step generation, the forward context is prioritized.
3. **RNN with Attention:** This model augments the vanilla RNN with an attention module that calculates alignment scores between the current hidden state and all previous states. This allows the model to "attend" to specific characters earlier in the name when predicting the next character.

1.0.2 Hyperparameters and Trainable Parameters

All models were trained for 50 epochs using the Adam optimizer with a learning rate of 0.001 and a hidden size of 128.

Model	Layers	Hidden Size	Learning Rate	Total Parameters
Vanilla RNN	1	128	0.001	23,836
RNN + Attention	1 + Attn	128	0.001	60,316
BLSTM	1 (Bidir)	128	0.001	168,988

Table 4: Model architecture specifications and parameter counts.

2 Task 2: Quantitative Evaluation

The models were evaluated on their ability to generate 1,000 names. We focus on **Novelty** (percentage of names not in the training set) and **Diversity** (unique names vs. total generated).

Model	Final Loss	Novelty Rate (%)	Diversity
Vanilla RNN	1.3194	50.00%	1.0000
RNN + Attention	1.3019	40.00%	1.0000
BLSTM	1.1196	10.00%	1.0000

Table 5: Quantitative comparison of generative performance.

3 Task 3: Qualitative Analysis

3.1 Realism and Sample Names

The **BLSTM** achieved the highest realism scores, generating names like *Umashankar* and *Lakshman*. However, its low novelty rate (10%) indicates it primarily memorized the training set. The **Vanilla RNN** and **Attention** models generated "pseudo-names" that are phonetically plausible but often non-existent in the training data, demonstrating higher creative novelty.

Model	**Representative Generated Samples**
Vanilla RNN	Reshma, Dini, Dived, Akshay, Naveen, Tarvi, Anupa, Naveek, Param, Nirav
RNN + Attention	Bhakti, Janak, Bhoomi, Shashika, Shakti, Somesh, Ramakatt, Dharmendra, Sovit
BLSTM	Umashankar, Nagaraj, Rakhi, Rohit, Shiva, Lakshman, Tathagata, Vidya

Table 6: Qualitative samples of generated Indian names.

3.2 Common Failure Modes

- **Memorization (BLSTM):** The model overfits to the most common sequences in the dataset, leading to low novelty.
- **Phonetic Hallucination (Vanilla RNN):** The model generates names like *Naveek* or *Dived*. While they follow the structural patterns of the training data, they are not actual names.
- **Sequence Termination Issues (Attention):** The attention model sometimes struggles to predict the `<EOS>` token at the right time, leading to slightly distorted endings like *Ramakatt* or *Addevivi*.